

Immersive Model for Lake Mead based on the Principle of Division of Reservoir Inflow

Let's Start!



UNIVERSITY OF CALIFORNIA
Agriculture and Natural Resources

David E. Rosenberg

david.rosenberg@usu.edu

<http://rosenberg.usu.edu>

Erik Porse

<https://ucanr.edu>

January 2025

Purpose

Immerse in and personify water user roles for a Lake Mead model based on the principle of divide reservoir inflow

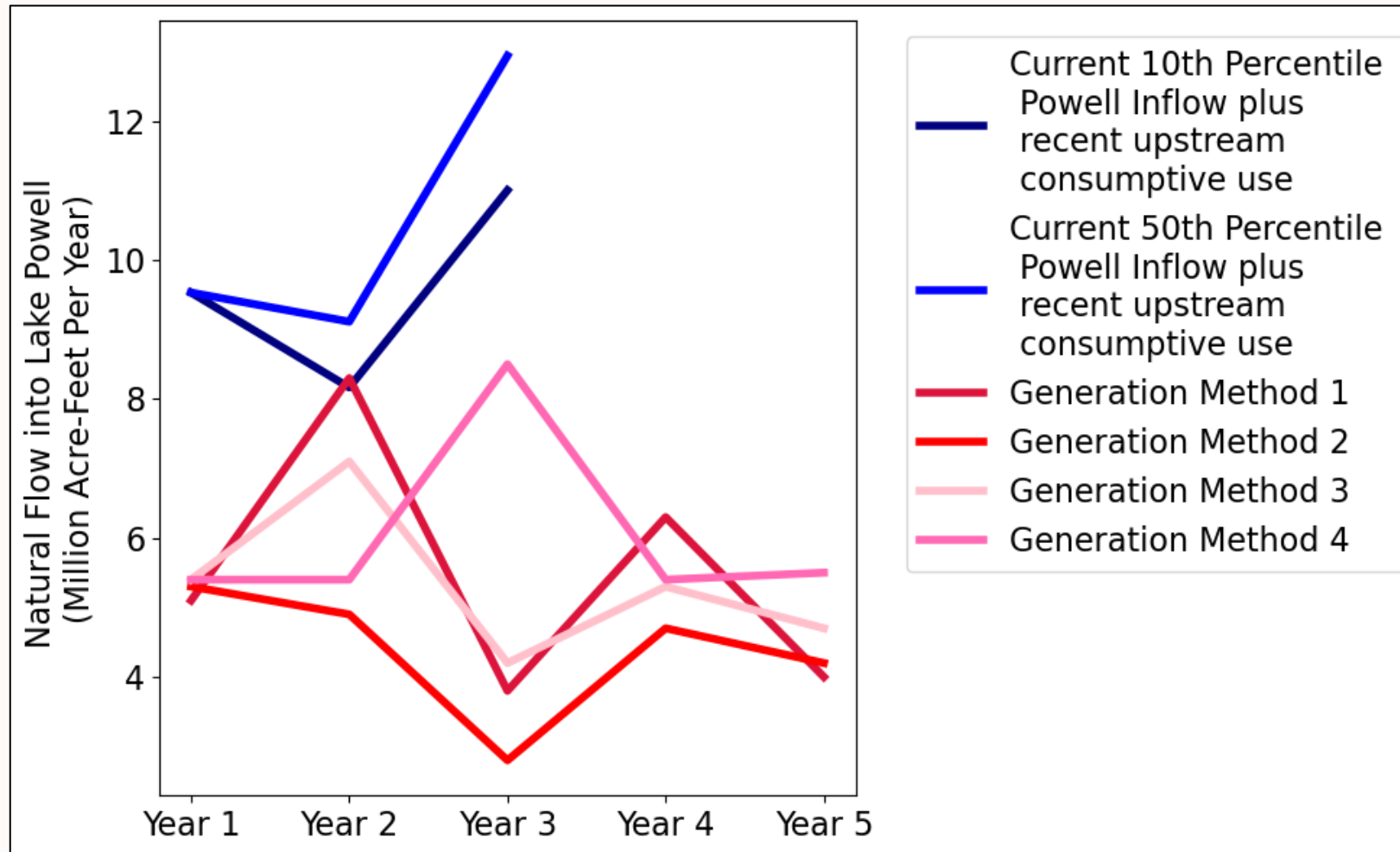
Learn why:

- Basin partners choose assumptions.
- Articulate strategies to manage risk.
- Modify strategies in response to their:
 - Available water.
 - Other's choices.
 - Real-time discussion of choices.

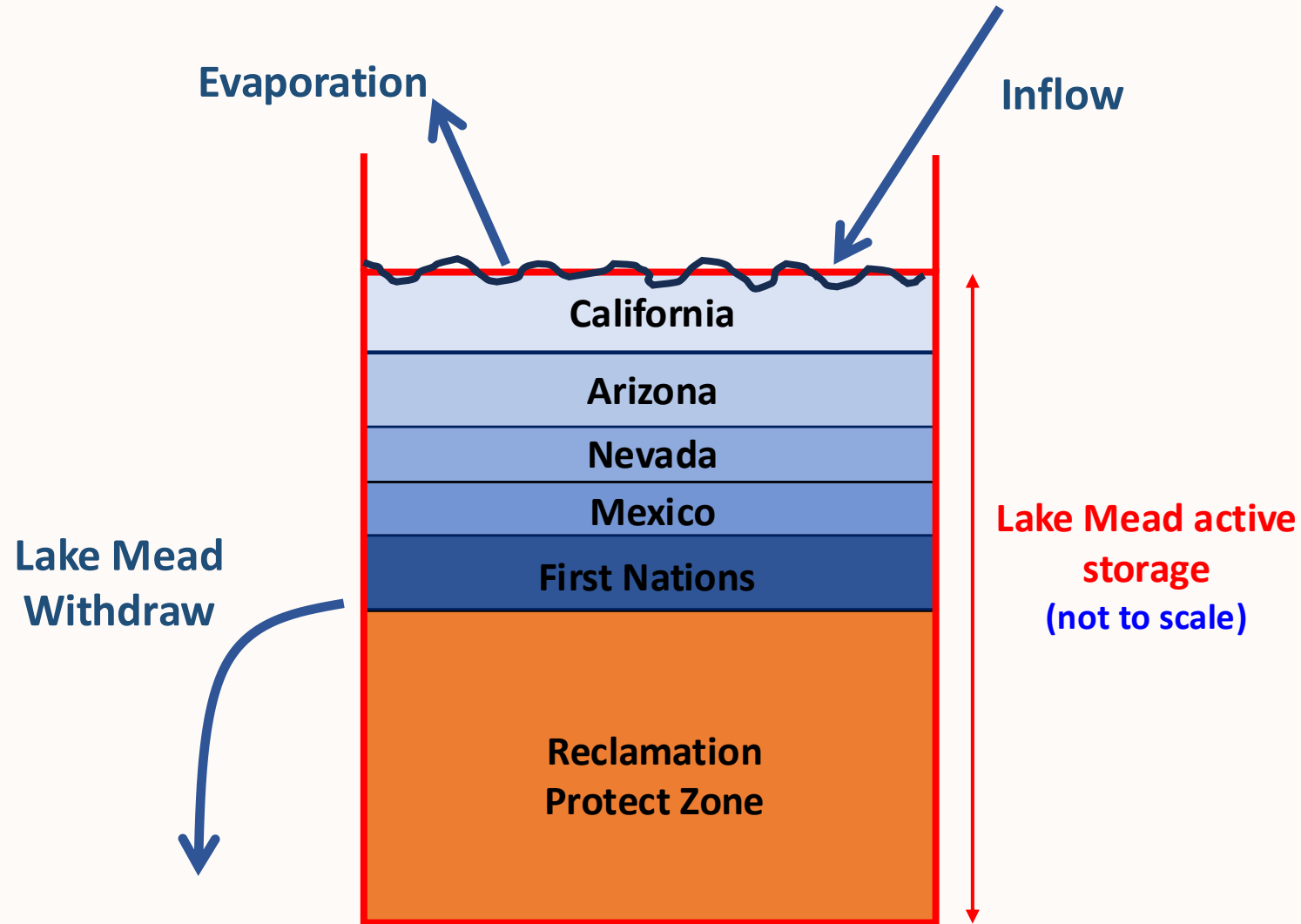
Provoke thought and discussion to:

- Stabilize and recover reservoir storage with low storage and low inflow.
- Increase user autonomy to manage their conflicting vulnerabilities to water shortages.

1. Colorado River flows will be more volatile, declining, and have longer-lasting periods of low flow



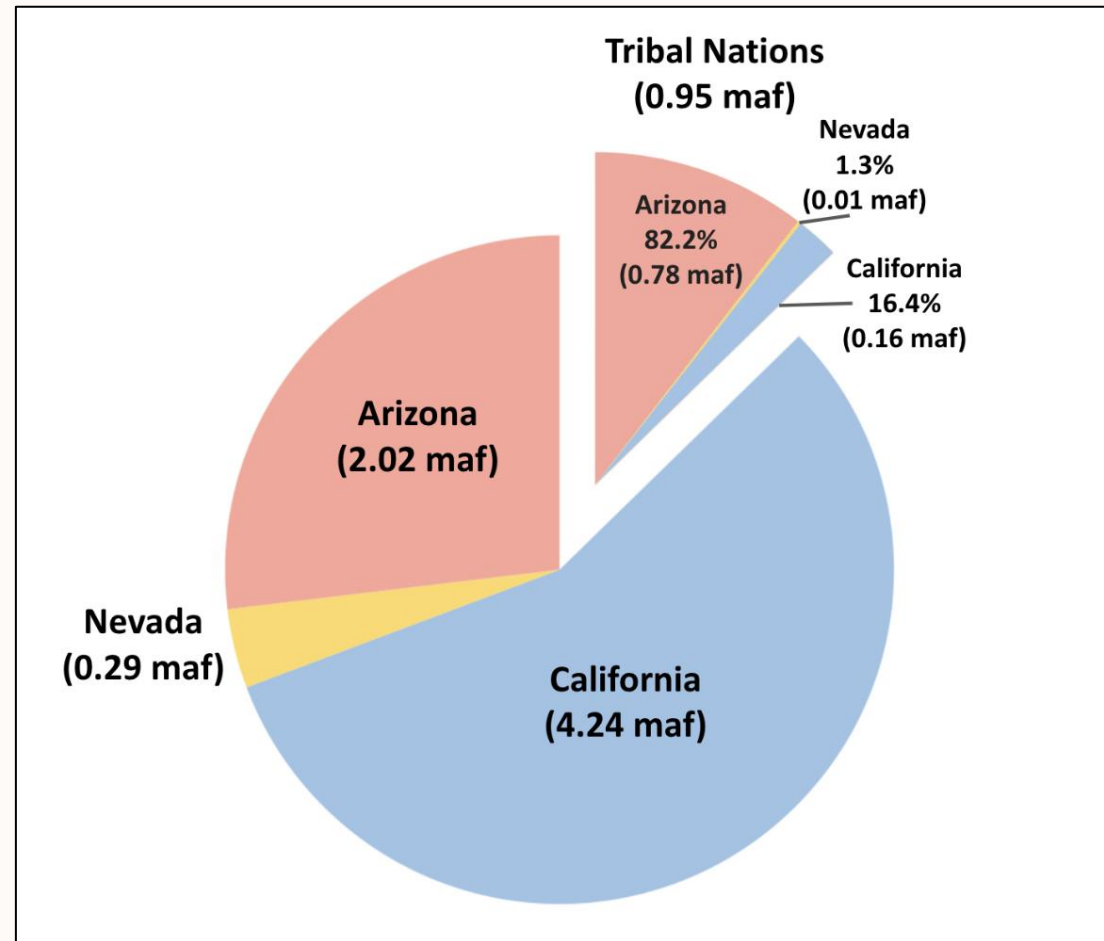
2. Lake Mead water level is the sum of the protection elevation and each user's available water



3. Each user manages all their available water not just prior conserved water.

$$\begin{array}{ccccccc} \text{Available} & & & & & & \\ \text{Water} & = & \text{Prior} & + & \text{Share of} & - & \text{Share of} & + & \text{Purchases} & - & \text{Sales} \\ & & \text{Available} & & \text{Lake} & & \text{Evaporation} & & & & \\ & & \text{Water} & & \text{Mead} & & & & & & \\ & & & & \text{inflow} & & & & & & \\ & & & & & & & & \underbrace{\hspace{10em}} & & \\ & & & & & & & & \text{Optional} & & \end{array}$$

4. Tribal Nations of the Lower Basin manage their own settled water rights



Ready to Immerse?



1. Identify session guide



2. Download (link or QR code)



3. Move into Google Drive



4. Invite participants

5. Open ReadMe-Directions sheet



<https://tinyurl.com/ImmerseLakeMead>

David E. Rosenberg

Utah State University

<http://rosenberg.usu.edu>

@WaterModeler